



increase accuracy for LED lighting and allow users to carefully control LED light output levels without sacrificing colour or light quality for reliable long-life applications. These intelligent

pulse width modulation controllers have fully integrated PIC MCU cores. The digital interface allows for communication and configuration, allowing a subsystem to report status or be remotely controlled.

Microchip Technology Inc.
www.microchip.com

Power supply IC

Renesas has released high-efficiency power supply IC, the RAA230161, that supports the requirements defined in USB Power Delivery (USB PD) 2.0 and 3.0 specifications. This IC can be used with equipment that uses DC power, from AC adaptors and PC peripheral equipment to home and office equipment, and hub applications.

The DC/DC converter block in the device achieves up to 95% efficiency,



leading to lower power loss. Its 20-pin TSSOP package with built-in heat-sink

suppresses the heat generated due to increased power levels. The step-down DC/DC converter handles currents up to 3A, operates at 500kHz, has an input voltage of 24V and output voltages of 5V, 9V, 12V, 15V and 20V.

The IC monitors the output current and protects the system so that current in excess of a set value is not output. It incorporates over-voltage protection, over-current protection and short-circuit protection.

Renesas Electronics Corporation
www.renesas.com

Space saver relay module

Shavison space-saver relay module has eight 1C/O relays. Its features include compact size, 35mm DIN rail mounting, free-wheeling diode across coil, LED indication for coil on/off and reverse-polarity-

blocking diode. The reverse-polarity-blocking diode prevents damage to PLC outputs in case of incorrect wiring. The module has a front cover that covers wiring and gives a decent look.

The conventional relay module is 150mm wide while this 8-channel 1C/O space-saver module is just 55mm wide. It is possible to mount eight space-saver relay modules in



the space occupied by three conventional relay modules.

Shavison Electronics Pot Ltd
www.shavison.com

DC/DC converter

MORNSUN recently released 15W and 40W power supply modules, PV15/PV40-29Bxx series, which provide 200-1500V DC input voltage, high isolation voltage of 4000V AC and can be used at high altitudes up to 5000m. In photovoltaic industry 1500V DC systems are being used in place of the 1000V DC systems. Higher-voltage systems enable longer strings, which allow for fewer combiner boxes, less wiring and trenching, and therefore less labour. The modules are designed with input under-voltage protection, which protects a system from frequent restart. The brand new PV15/PV40-29Bxx series are suitable for various applications such as photovoltaic combiner, photovoltaic inverter, high voltage switching, etc for operation from -40° to +70°. These meet EN62109 standard, have high efficiency and produce low ripple and noise.

Mornsun Guangzhou Science & Technology Co., Ltd; www.mornsun-power.com

MISCELLANEOUS

IP Phone

Matrix presents Sparsh VP510E, an IP phone with context-sensitive keys that change with the requirement. It comes with 16DSS keys, busy lamp field and built-in PoE offering pro-

ductivity features for faster operations and collaboration. Has 240 x 64 pixel graphical LCD with backlight, 4 context-sensitive keys, fixed function keys (with LED), message waiting and ringer lamp, headset interface, full duplex speakerphone, PC and LAN Ethernet ports and power over Ethernet.



Matrix Comsec
www.MatrixComsec.com

Load development kit

Vicor Corporation has launched a new AC to point-of-load development kit (ACPoL DKit). It enables power system designers to immediately begin operating and evaluating a complete power system solution without the hassles and time lag associated with designing and acquiring their own prototyping boards, system interconnects, energy storage and thermal management components. One version of the DKit features a 24V DC distribution bus; another version features a 48V DC bus. DKits include all necessary ancillary components, including a bus hold-up capacitor assembly and an AC line cord.

Vicor Corporation
www.vicorpower.com

Bluetooth thermal printer

DYNO_BT2A light-weight thermal printer, which prints on 5cm thermal paper, provides fast and low-cost printing in mobile and wireless environments.



Its specifications include Bluetooth 4.0, USB device, serial UART, configurable baud rate, NFC (optional), thermal line printing, hand tearing, 50km print-head core life, two English fonts, regional language, graphics, Bar and QR codes.

It can accommodate up to 9 images, has 7.4V, 1200/1400mAh battery with on-board battery charge management, a 9V (1A) charger, gives various indications, and supports Bluetooth connectivity via smartphone, PDA, laptop, netbook, etc.

Coinel Technology Solutions LLP
www.coineltech.com